

Bilal Amin

+49 176 86534249 | write2bilalamin@gmail.com | linkedin.com/in/aminbilal
github.com/Astavicious | bilal-amin-portfolio-eight.vercel.app | Saarbrücken, Germany

PROFILE

Master's student in Data Science and AI with a background in mechatronics and hands-on experience spanning network simulation, data analysis, web development, and LLM applications. Interested in applied AI, intelligent software systems, and building data-driven products from prototype to deployment.

EXPERIENCE

- Research Assistant (HiWi) – Software Engineering** Oct. 2026 – Present
Chair of Software Engineering, Saarland University Saarbrücken, Germany
 - Will work directly with Qingqing Dong on AI-assisted software engineering and model-driven system engineering research.
 - Planned work includes investigating AI-assisted model merging, semantic merge-conflict detection, and feature-based merge recommendations.
 - Will support research on software evolution and intelligent methods for improving complex software and model-based systems.
- Student Assistant (HiWi) – Web Development** Sep. 2026 – Present
EuroComCenter, Saarland University Saarbrücken, Germany
 - Maintain and improve a multilingual learning platform covering 6 Romance languages and structured around the 7-Sieves intercomprehension methodology.
 - Reviewed and updated 10+ CMS-managed pages and content items, resolving technical, layout, and user-interface issues across the platform.
 - Translate project requirements into practical improvements to navigation, content structure, usability, and multilingual presentation.
- Working Student – Internet Architecture Group** Oct. 2024 – Mar. 2025
Max Planck Institute for Informatics (MPI) Saarbrücken, Germany
 - Designed an automated NS-3 evaluation framework comparing 5 congestion-control algorithms across 10 network scenarios each, covering 50 algorithm-scenario evaluations.
 - Built a dashboard visualizing 4 key metrics – throughput, RTT/latency, packet loss, and congestion window – across experiment configurations.
 - Developed 5 automated system tests and reusable evaluation logic to support repeatable performance experiments.
 - Analyzed simulation results to identify algorithm-specific behavior and guide experiment and design decisions.
- Data Analyst** Oct. 2023 – Mar. 2024
Convergent Business Technologies Islamabad, Pakistan
 - Completed structured training in programming, data science, and business analytics, applying C, Python, and SQL through hands-on assignments and mentor evaluations.
 - Built 5 Tableau/Power BI dashboards and presented analytical findings to technical and business stakeholders in simulated client settings.
 - Delivered an end-to-end Employee Attrition Prediction capstone using approximately 600 employee records, covering data preparation, modeling, evaluation, and presentation.
 - Built a business-focused Dognition analytics dashboard and gained practical exposure to dimensional modeling, ETL concepts, SQL optimization, and Agile/Scrum workflows.
- Web Development Intern** Jun. 2023 – Aug. 2023
CureMD Lahore, Pakistan
 - Built or significantly modified 18 responsive UI pages/components using HTML, CSS, and JavaScript for web-based healthcare software.
 - Implemented approximately 6 .NET API endpoints to support application data exchange, backend integration, and internal tool development.
- Technical Intern** Aug. 2021 – Oct. 2021
STEP (Space Technology Education and Popularization) Islamabad, Pakistan
 - Led 18 hands-on Scratch programming workshops – 6 larger and 12 smaller sessions – typically serving groups of 50–70 students.
 - Created beginner-friendly educational material and practical activities to make programming and STEM concepts accessible to young learners.

EDUCATION

Universität des Saarlandes

Master's degree, Data Science and AI – Grade: 1.9

University of Bergen – Erasmus+ Exchange Semester

Exchange Student, Computer Science

National University of Sciences & Technology

Bachelor's in Mechatronics Engineering (Computer Science) – Grade: 1.7

Apr. 2024 – Present

Saarbrücken, Germany

Jan. 2026 – Jun. 2026

Bergen, Norway

Sep. 2019 – Sep. 2023

Islamabad, Pakistan

PROJECTS

GermanLyft – AI-Powered German Learning Platform

2026

- Developed a full-stack German-learning platform for 2 primary CEFR levels (A1–A2) using React, TypeScript, and Flask.
- Implemented 5 major user-facing features: placement tests, quizzes, user accounts, profiles, and Anki .apkg deck importing.
- Built 4 exercise formats – multiple choice, fill-in-the-blank, sentence correction, and vocabulary matching – exposed through approximately 3 backend API endpoints.
- Tested the platform with 3 users and deployed the frontend on Vercel and backend on Render as 2 independently hosted services.

AI Learning Assistant using LLMs (RAG)

2025

- Built an AI learning assistant with Python, LangChain, Streamlit, and FastAPI using a locally hosted Llama model.
- Tested retrieval and question answering across 25 Functional Programming course documents – 13 lecture PDFs and 12 worksheets.
- Evaluated the assistant with 12+ questions per document, representing 300+ question-document interactions across the test corpus.
- Implemented PDF processing, vector embeddings, ChromaDB retrieval, citation-based answers, summaries, quizzes, and study-material generation.

Smart StudyDesk: AI-Powered Study Organizer

2025

- Designed a cross-platform study application with 4 core learning features: AI note summaries, personalized quizzes, progress tracking, and weak-area analysis.
- Conducted usability testing with 4 participants using semi-structured interviews and NASA-TLX workload analysis.
- Identified approximately 5 usability issues and iterated on navigation, folder organization, and feedback visibility based on participant feedback.

Side Hustle Launchpad – Tech4Europe Hackathon, KIT

2025

- Built a full-stack small-business planning application in a 48-hour Tech4Europe Hackathon at Karlsruhe Institute of Technology as part of a 4-person team.
- Owned most backend and database work using FastAPI and MongoDB, while also contributing to the pitch deck and final presentation.

MarketPulse Website Design and HCI Analysis

2024

- Designed and developed 5 responsive MarketPulse pages/screens using HTML, CSS, and JavaScript.
- Applied 6 HCI principles – consistency, simplicity, feedback, visibility, constraints, and accessibility – and tested the interface with 3 users.

TECHNICAL SKILLS AND INTERESTS

Programming : Python, JavaScript, TypeScript, SQL, HTML, CSS, C++, C#, Java, Rust

Machine Learning & AI : Scikit-Learn, PyTorch, TensorFlow, Hugging Face Transformers, LLMs, Prompt Engineering, RAG

LLM Frameworks : LangChain, LangGraph, Ollama, ChromaDB

Web Frameworks : FastAPI, Flask, Streamlit, ReactJS

Libraries & Data : Pandas, NumPy, PySpark, Matplotlib

Tools & Platforms : Git, GitHub Actions, Docker, Microsoft Azure, VS Code, Jupyter, Power BI, Tableau, Figma

Interests : LLM Applications, Agentic AI, AI-Assisted Software Engineering, Data Engineering, Workflow Automation